

Kubernetes Ingress Not Routing Traffic SOP

Objective

Restore HTTP or HTTPS routing when Kubernetes ingress rules do not route traffic to the expected backend service.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T017
Issue	Kubernetes ingress not routing traffic
Category	Networking
Service	Kubernetes
Severity	High
Status	Resolved
Description	Traffic not reaching service
Expected Resolution	Check ingress rules and controller logs

Procedure

1. Confirm impacted hostname, path, HTTP status code, and ingress name.
2. Review ingress rules for host, path, pathType, service name, and service port.
3. Check ingress controller pod status, logs, and recent reload errors.
4. Validate TLS secret, certificate validity, and DNS resolution for the host.
5. Confirm backend service has healthy endpoints and correct port mapping.
6. Test service connectivity directly from inside the cluster.
7. Update ingress rule, TLS secret, DNS, or backend service configuration.
8. Retest routing and monitor ingress controller errors.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Traffic not reaching service	Wrong host or path rule	Correct ingress rule
502 or 503 error	No healthy backend endpoints	Fix service endpoints or pod readiness
TLS error	Invalid or missing certificate secret	Update TLS secret
Ingress ignored	Controller class mismatch	Set correct ingressClassName

Best Practices

1. Use explicit ingressClassName.
2. Validate DNS and TLS before release.
3. Monitor ingress controller error rate.
4. Keep backend service and ingress ports aligned.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)